

Week 5 — Session 1

Course: 10211CS225 — Problem Solving and Testing using Java

Unit II: Algorithmic Thinking & Competitive Problem Patterns

Topic: Maximum Subarray (Brute → Optimized)

Language: Java

Focus: Use linear-time scanning and a sliding window to optimize common subarray tasks.

Task 1: Maximum Subarray

Practice: <https://leetcode.com/problems/maximum-subarray/>

Complete Java Solution

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] a = new int[n];
        for (int i=0; i<n; i++) a[i]=sc.nextInt();

        int current=a[0], best=a[0];
        for (int i=1; i<n; i++) {
            current=Math.max(a[i], current+a[i]);
            best=Math.max(best, current);
        }
        System.out.println(best);
        sc.close();
    }
}
```

Task 2: The Birthday Bar

Practice: <https://www.hackerrank.com/challenges/the-birthday-bar/problem>

Complete Java Solution

```
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int[] a=new int[n];
        for(int i=0; i<n; i++) a[i]=sc.nextInt();
        int d=sc.nextInt(), m=sc.nextInt();

        int sum=0, count=0;
        for(int i=0; i<n; i++){
            sum+=a[i];
            if(i>=d) sum-=a[i-d];
            if(i>=d-1 && sum==m) count++;
        }
        System.out.println(count);
        sc.close();
    }
}
```

Programs are standalone Java solutions. Java 17+ is recommended.